

Regression Analysis Report on Car Prices 📊🚗

1. Introduction

This report presents an Ordinary Least Squares (OLS) regression analysis to determine the factors influencing car prices. The objective is to identify the key predictors of car price variations and analyze their statistical significance.

2. Dataset Description

The dataset consists of 10,000 car records with the following variables:

- **Price:** The dependent variable (car price in USD).
- **Engine_Size:** The size of the engine in cc.
- **Mileage:** The distance the car has traveled (in kilometers).
- **Doors:** The number of doors in the car.
- **Owner_Count:** The number of previous owners.

```
df=df[['Engine_Size','Mileage','Doors','Owner_Count','Price']]
```

```
df.head()
```

	Engine_Size	Mileage	Doors	Owner_Count	Price
0	4.2	289944	3	5	8501
1	2.0	5356	2	3	12092
2	4.2	231440	4	2	11171
3	2.0	160971	2	1	11780
4	2.6	286618	3	3	2867

OLS Regression Results						
=====						
Dep. Variable:	Price	R-squared:	0.438			
Model:	OLS	Adj. R-squared:	0.438			
Method:	Least Squares	F-statistic:	1945.			
Date:	Tue, 11 Feb 2025	Prob (F-statistic):	0.00			
Time:	23:06:29	Log-Likelihood:	-91742.			
No. Observations:	10000	AIC:	1.835e+05			
Df Residuals:	9995	BIC:	1.835e+05			
Df Model:	4					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

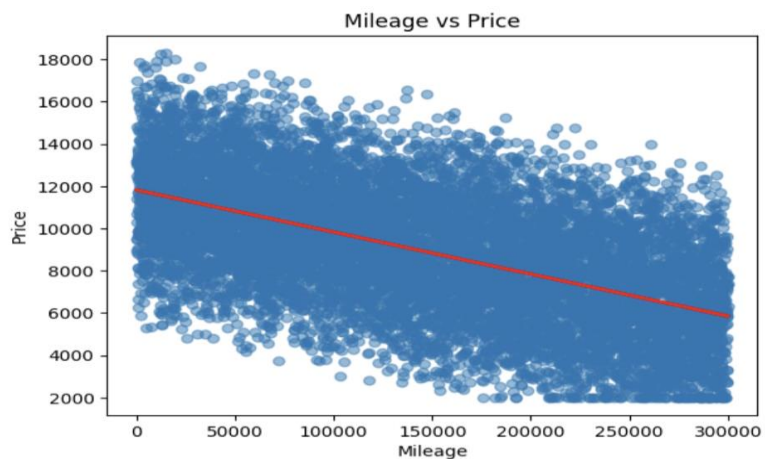
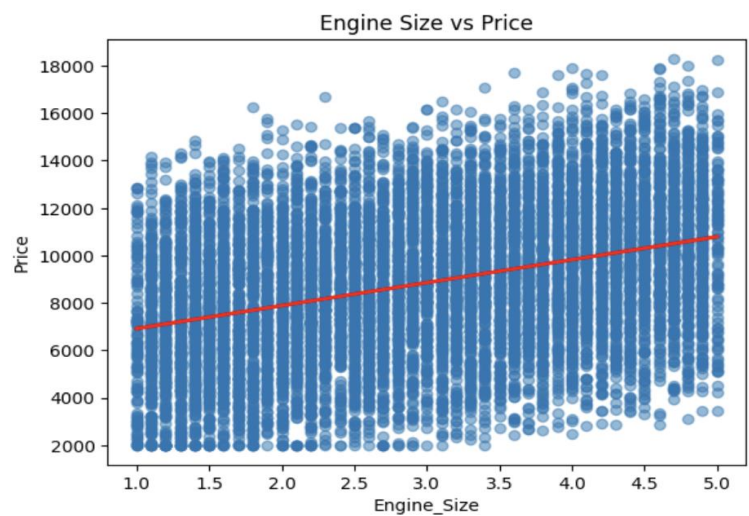
Engine_Size	990.9018	20.316	48.774	0.000	951.078	1030.725
Mileage	-0.0201	0.000	-74.232	0.000	-0.021	-0.020
Doors	23.8897	21.032	1.136	0.256	-17.337	65.117
Owner_Count	7.8710	16.410	0.480	0.631	-24.297	40.039
const	8769.1844	116.828	75.060	0.000	8540.177	8998.191
=====						
Omnibus:	732.610	Durbin-Watson:	1.973			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	250.648			
Skew:	0.054	Prob(JB):	3.74e-55			
Kurtosis:	2.232	Cond. No.	8.73e+05			
=====						

4. Data Visualization

- **Scatter Plot:** Engine Size vs. Price
- **Scatter Plot:** Mileage vs. Price

```
# Engine Size and Price
sns.regplot(x=df["Engine_Size"], y=df["Price"], scatter_kws={"alpha":0.5}, line_kws={"color":"red"})
plt.title("Engine Size vs Price")
plt.show()

# Mileage and Price
sns.regplot(x=df["Mileage"], y=df["Price"], scatter_kws={"alpha":0.5}, line_kws={"color":"red"})
plt.title("Mileage vs Price")
plt.show()
```

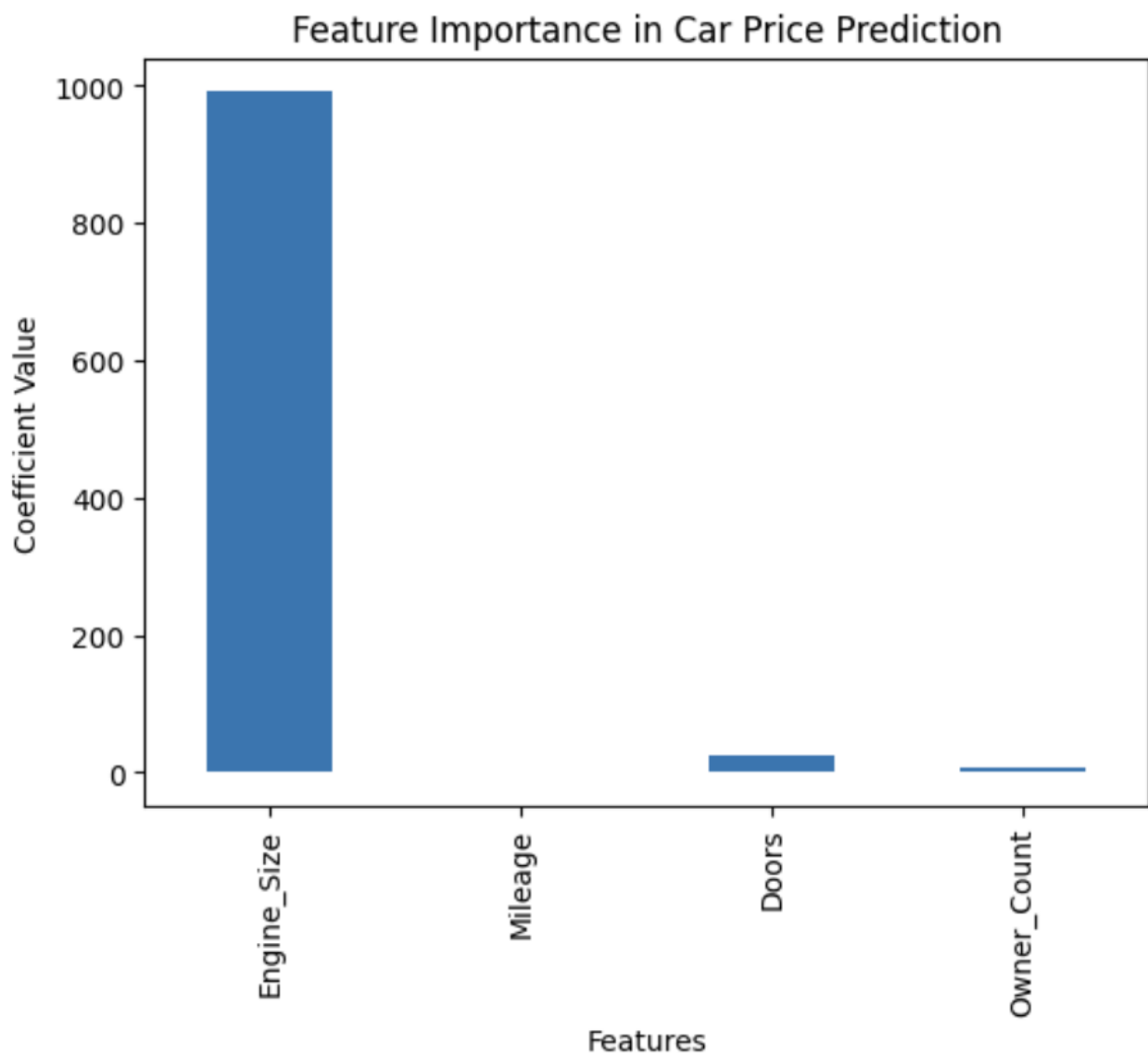


- **Bar Chart:** Feature Importance (Regression Coefficients)

```
import pandas as pd

coef_df = pd.DataFrame({"Feature": ["Engine_Size", "Mileage", "Doors", "Owner_Count"],
                        "Coefficient": [990.9018, -0.0201, 23.8897, 7.8710]})

coef_df.plot(x="Feature", y="Coefficient", kind="bar", legend=False)
plt.title("Feature Importance in Car Price Prediction")
plt.xlabel("Features")
plt.ylabel("Coefficient Value")
plt.show()
```



5. Conclusion & Recommendations

- The model suggests that **Engine Size and Mileage are the most significant factors** affecting car prices.
- The low impact of Doors and Owner Count indicates that buyers do not strongly consider these features when determining car value.
- To improve the model, additional features like **Brand, Model, Year, Fuel Type** should be included.

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